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ORIGINAL RESEARCH

Neuroanatomical substrates of maximizing tendency in decision-making: a voxel-based morphometric study

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Abstract

Maximizing tendency is a central decision-making concept that has increasingly drawn attention from the scientific community. It refers to individuals' predisposition to look for the best option instead of settling for something that merely passes an internal threshold of acceptability. Although this maximizing strategy intuitively increases individual benefits, it also has been linked to various negative outcomes, including decreased well-being and low life satisfaction, and it varies considerably across populations. However, the neuroanatomical characteristics underlying this heterogeneity remain poorly understood. To address this knowledge gap, a 13-item Maximization Scale and magnetic resonance imaging technique were respectively used in this study to estimate individual maximizing tendency and structural morphological information on a sample of healthy adults (n = 69). Furthermore, voxel-based morphometry (VBM) analysis was conducted to investigate the associations between gray matter volume (GMV) and maximizing tendency through univariate and multivariate pattern analysis (MVPA). Univariate analysis did not determine an association between maximizing tendency and whole-brain GMV; by contrast, MVPA revealed that maximizing tendency could be successfully predicted by the GMVs of the right inferior frontal gyrus (IFG), right insula, and right cerebellum. These findings suggest the critical role of the morphological characteristics of the cortical-subcortical circuitry in individuals' maximizing tendency.

Keywords Decision-making · Maximizing tendency · Multivariate pattern analysis · Voxel-based morphometry

Introduction

Classical economic theory assumes that human beings are always rational and pursue the maximization of utility or rewards(von Neumann & Morgenstern, 1944). Maximization has thus become a central concept in economic decision-making. Simon (1955) proposed that people cannot fully realize maximization due to the constraints of human cognitive abilities and the complexity of decision-making situations. These constraints cause substantial individual differences in maximization when people make decisions in everyday contexts, such as choosing a college major (Leach & Patall, 2013) or choosing a friend (Newman et al., 2018). Some people tend to satisfice and settle for options that are simply good enough, whereas others consistently pursue the best option. To precisely capture these individual differences, Schwartz et al., (2002) developed the Maximization Scale (MS) for assessing maximization tendency on the basis of seminal theory proposed by Simon (1955, 1956). People who tend to use high standards to make the best decision only after they have evaluated all alternatives are defined as "maximizers." By contrast, people who tend to select a "good enough" option that meets a minimum threshold of acceptability are satisficers. In terms of the MS, several researchers have attempted to explore the psychological mechanism of maximization (e.g., Bruin et al., 2015; Iyengar et al., 2006; Luan & Li, 2017). However, knowledge of the underlying structural neural substrates of maximizing tendency remains scant.

Maximizing tendency is multifaceted and depends on three sub-dimensions: alternative search, high standards, and decision difficulty (College et al., 2008; Schwartz et

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al., 2002). The first sub-dimension, alternative search, represents the tendency to seek better options. Because a maximizer seeks to obtain the best option, they are willing to sacrifice resources to search for more options. With more options available, they are more likely to make a nonoptimal choice, and this prospect may undermine whatever pleasure the decision-maker obtains from their actual choice (Schwartz et al., 2002). Researchers have proposed a notion called the "Maximization Paradox," a phenomenon wherein maximizers tend to sacrifice resources such as time and cognitive effort to attain more options, which ultimately decreases their satisfaction and increases other negative outcomes (Dar-nimrod et al., 2009). This notion has been supported by several studies, which have linked the maximizing tendency to various detrimental psychological effects, such as higher levels of depression, neuroticism (Diab et al., 2008; Purvis et al., 2011), and regret (Schwartz et al., 2002) and lower levels of happiness, optimism, self-esteem (Rim et al., 2011), well-being (Vargová et al., 2020), and life satisfaction (Moyano-díaz, 2014). Therefore, maximizing tendency might largely be associated with emotional components (i.e., negative emotions).

The second sub-dimension, high standards, represents decision-makers' tendency to hold high standards for themselves and things in general. A maximizer searches for the "best" outcome, but the satisficer looks for something "good enough." Therefore, with the "best" as a criterion rather than "good enough," the maximizer is more inclined to experience regret if they discover that an option better than the chosen one was available (Schwartz et al., 2002). Furthermore, when the maximizer ends the search and chooses an option, they retain a lingering doubt that they could have done better by searching a bit more. Therefore, the maximizer demonstrates more counterfactual thinking. Moreover, in general, the maximizer lacks sufficient information to make an unambiguous judgment. Although what is "good enough" can usually be judged in absolute terms, what is the "best" may often require social comparison. Therefore, the maximizer may be more concerned with their relative position, and their decision-making may be dependent on others (Wandi Bruine De Bruin et al., 2007). However, whether such explanation of counterfactual thinking and upward social comparisons can be validated on the basis of the brain's morphological characteristics remains unknown.

The third sub-dimension, decision difficulty, reflects the difficulty associated with making decisions. Maximizers tend to search for more options; however, searching for more options would create a seemingly intractable information problem (Schwartz et al., 2002) that might increase the difficulty of decision-making and reduce the maximizers' confidence in decision-making. Furthermore, maximizers highly depend on external information sources, which might further induce them to distrust their finalized choices (Iyengar et al., 2006). Rim (2017) observed that individuals' maximization scores were negatively related to their confidence ratings.

As a critical one of demographical variables, sex effects have been considered as an important topic in the domain of decision-making. For example, males are found larger choice impulsivity(Lv et al., 2020), more concerning about fairness(Saad & Gill, 2001), and higher risk taking (Powell & Ansic, 1997; Schmidt et al., 2015). Furthermore, such sex differences were also identified in the social comparisons (Guimond et al., 2006), manifesting as greater degree of uniqueness bias across both physical attributes and socially-desirable personality characteristics in men rather than in women (Furnham & Dowsett, 1993). In addition, the evidence from sex differences in regret (Roese et al., 2006) and in counterfactual thinking(Awo et al., 2019) has been provided in prior empirical studies. However, the sex differences on the maximizing are mixed, manifesting that some observed the sex effects(Schwartz et al., 2002) while others highlighted gender invariance in maximization(Richardson et al., 2014).

In the past decade, functional magnetic resonance imaging (fMRI) has broadly become an essential tool for understanding how the human brain works, and it provides a novel window for exploring the structural architectures and functional organizations of the human brain and their associations with specific cognitive processes and decision-making strategies (Fellows, 2004). Although numerous studies have focused on the brain–behavior associations from the perspective of individual differences in decision-making, the neural substrates underlying individual differences in maximizing tendency still remain unexplored. The literature indicates that the psychological processes of maximization might largely rely on cognitive (i.e., upward social comparisons, counterfactual thinking, and confidence) and emotional components (i.e., regret). Some imaging studies have implicated the engagement of several brain regions on these components, including the lateral prefrontal cortex, right inferior frontal gyrus (IFG), medial orbitofrontal cortex, insula, and cerebellum (Lindner et al., 2015; Van Hoeck et al., 2012). Thus, we hypothesized that the structural characteristics of these cortical areas might be associated with individual differences in maximizing tendency.

The main objective of this study was to investigate the potential neuroanatomical substrates of maximizing tendency by using both univariate analysis and multivariate pattern analysis (MVPA). Gray matter volume (GMV) is a critical index that represents the morphological attributes of the brain and enables a voxel-wise estimation of the local amount of a specific tissue (Kurth et al., 2015). Accordingly, this index was used in the present study. In addition, compared with univariate analysis, MVPA has been demonstrated to exhibit greater sensitivity to potential brain–behavior associations and to be more predictive of behavioral performance in decision-making (Huang et al., 2019; Jimura & Poldrack, 2012; Wang et al., 2014, 2016). Thus, by combining these advantages, we comprehensively investigated the possible associations between GMV and individual differences in maximization scores.

Materials and methods

Participants

Sixty-nine healthy adult participants (33 men; aged 21.59 ± 2.81 years) were recruited through a university bulletin board system and advertisements. No volunteer was excluded from the study due to low imaging quality. All participants had normal or corrected-to-normal vision and reported no neurological or psychiatric history. Informed written consent was obtained from participants before formal experiments were conducted. This study was approved by the Institutional Review Board of the School of Psychological and Cognitive Sciences at Peking University and Faculty of Psychology at Tianjin Normal University.

Questionnaire

In this study, MS—which consists of three facets (i.e., alternative search, decision difficulty, and high standards)—was implemented to measure individual differences in maximizing tendency during decision-making (Schwartz et al., 2002). The alternative search facet scale comprises six items and measures the degree to which an individual sustains their search for "better" alternatives, even after having found a satisfying option. The decision difficulty facet comprises four items and measures the difficulty experienced in everyday decision-making. Finally, the high standards facet comprises three items and measures the degree to which the individual is satisfied only by meeting the highest standards and choosing the single best option. For each of these facets, each item was rated on a 7-point Likert-type scale with anchors ranging from 1 ("strongly disagree") to 7 ("strongly agree"). The total scores ranged from 13 to 91, with a higher score indicating a greater maximizing tendency during decision-making.

Brain imaging data acquisition

Whole-brain imaging data were collected using the General Electric Signa VH/i 3.0T MRI scanner at the Center for MRI Research at Peking University. Participants lay supine on the scanner bed. Foam pads were utilized to minimize head movement. A T1-weighted three-dimensional gradient-echo pulse-sequence (MPRAGE) was used to acquire anatomical imaging data. The following parameters were adopted: TR = 2,560 ms; TE = 3.39 ms; flip angle = 7°; field of view = 256×256 mm^2; and voxel size = $1 \times 1 \times 1$ mm^3.

Structural MRI preprocessing

Structural MRI data were analyzed using, the Oxford Centre for Functional MRI of the Brain (FMRIB) Software Library (FSL)–voxel-based morphometry (VBM) toolbox. Brain images were extracted from the structural images, segmented according to tissue type, and then aligned to the gray matter template in the MNI152 standard space. Subsequently, the spatially normalized images were averaged to create a study-specific template, to which the native gray matter images were again registered using linear and nonlinear algorithms. The registered partial volume images were then modulated by dividing them with the Jacobian of the warp field to correct for local expansion or contraction. The modulated segmented images, which represented the GMV data, were then smoothed with an isotropic Gaussian kernel with a 3-mm standard deviation. Notably, nonsmoothed GMV data were further used for MVPA.

Univariate analysis

We examined associations between maximization scores and GMV of the whole brain by using a mixed-effects FLAME 1 model implemented in FSL. Age at MRI scan and total GMV were included as covariates.

In regression analysis, covariates were entered into the first block of equations. In the second block, mean-centered maximization scores and sex were entered. The interaction term, defined as the product of mean-centered optimization and sex, was entered into the third block. When the interactive effect was not significant, a reduced model, controlling for the same covariates and sex, was used to examine maximization in relation to the same outcome measures. Statistical results were determined at a cluster level ($Z > 2.3$, $p < 0.05$) and at a family-wise error rate of 0.05 to correct for multiple comparisons using Gaussian random field theory. We also examined the associations between GMV and different subdimensions of maximization (i.e., decision difficulty, alternative search, and high standards) by using a similar statistical model.

MVPA procedure

Epsilon-insensitive support vector regression (SVR; Drucker et al., 1997) with a linear kernel, as implemented in PyMVPA (http://www.pymvpa.org; Hanke et al., 2009), was used to examine the association between the maximization of decision-making and GMV in the whole brain. A searchlight procedure with a three-voxel radius (Kriegeskorte et al., 2006) was executed to provide a measure of decoding accuracy in the neighborhood of each voxel. On the basis of parameter settings in previous studies (Jimura & Poldrack, 2012; Wang et al., 2014), we set the ε parameter in the SVR to 0.01.

A three-fold cross-validation was applied in this study (Cui et al., 2018; Wang et al., 2016). The 69 participants were divided into 3 groups of 33 participants each, who were matched according to sex and maximization scores. In each iteration, an SVR model was trained on the basis of two groups and then applied to test the generalization of the remaining group on the basis of their imaging data. Notably, the training dataset was first normalized (i.e., mean subtracted out and then divided by the standard deviation), and the derived parameters were applied to normalize the testing dataset. Subsequently, the voxel-wise accuracy of the SVR predictions was calculated by deriving the Pearson's correlation coefficient between actual and predicted maximization values and then transformed to the corresponding *z* score maps. Finally, the SVR predictions were thresholded using cluster detection statistics, with a height threshold of $Z > 2.3$ and a cluster probability of $p < 0.05$, corrected for whole-brain multiple comparisons using Gaussian random field theory. A similar analysis strategy was performed to examine whether GMV could predict scores of the three subdimensions.

Correlational analysis

To directly compare the effects of univariate and multivariate statistics through Pearson's correlational analysis, we extracted the *z*-score of each voxel in regions of interest (ROIs) demonstrating significant predictions in MVPA and in the univariate analysis.

Results

Demographics

The maximization scores of decision-making varied from 30 to 91 (mean ± standard deviation = 57.77 ± 13.93). The scores in the alternative search, decision difficulty, and high standards dimensions were 25.06 ± 7.77, 18.75± 5.44, and

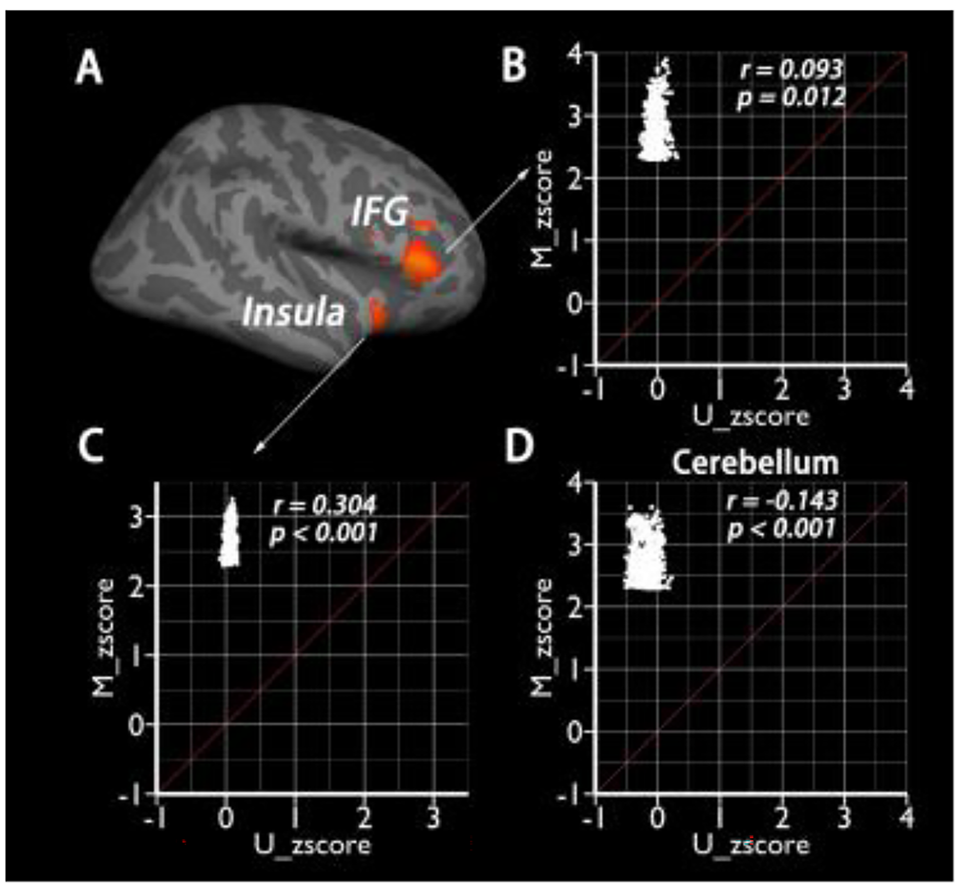

Fig. 1 MVPA results. (A) Individual differences in maximization could be successfully predicted by the GMV data in the right IFG, right insula and right cerebellum; (B-D) The z-score of univariate analysis (M_zscore) and MVPA (U_zscore) were significantly correlated in the IFG, insula and cerebellum. The scatter dots above diagonal line suggesting that the effect size was higher for MVPA than for univariate analysis

13.96 ± 3.72, respectively. We did not observe sex differences in the overall ratings of maximization ($t_{(67)} = -0.800$, $p = 0.427$) or the alternative search ($t_{(67)} = -1.053$, $p = 0.296$), decision difficulty ($t_{(67)} = -0.050$, $p = 0.961$), or high standards dimension ($t_{(67)} = -0.878$, $p = 0.383$). The scores of maximization and its three subdimensions did not vary significantly as a function of age (maximization: $r = -0.053$, $p = 0.667$; alternative search: $r = -0.059$, $p = 0.628$; decision difficulty: $r = -0.079$, $p = 0.521$; high standards: $r = 0.042$, $p = 0.733$) or total GMV (maximization: $r = 0.027$, $p = 0.826$; alternative search: $r = 0.026$, $p = 0.831$; decision difficulty: $r = -0.021$, $p = 0.862$; high standards: $r = 0.078$, $p = 0.525$). A Kolmogorov–Smirnov test revealed normal distributions of the maximization ($p = 0.450$), alternative search ($p = 0.972$), decision difficulty ($p = 0.682$), and high standards ($p = 0.543$) scores.

VBM results

First, by using a traditional univariate analysis, we explored whether an interaction effect existed between sex and maximization scores on the prediction of GMV. One major reason for this exploration is that previous research reported gender differences in maximization scores (Schwartz et al., 2002). However, our behavioral measures did not reveal this effect. Specifically, the interaction between sex and maximization did not affect the prediction of GMV during the whole-brain analysis. Moreover, the reduced model did not reveal a main effect of maximization on the prediction of GMV, even after age, sex, and total GMV were controlled for. We also did not obtain significant findings in the scores of any of the three facets of maximizing tendency.

Second, previous studies have demonstrated MVPA to be more sensitive than traditional univariate analysis in detecting brain–behavior associations (Jimura & Poldrack, 2012; Wang et al., 2014, 2016). Accordingly, we used this approach to examine the associations between maximizing tendency and the brain morphological index. As expected, MVPA revealed that individual differences in maximization could be successfully predicted by the GMV data in the right IFG (peak MNI = 60, 30, 6, $Z = 3.90$; cluster size = 741), right insula (MNI = 34, 22, − 8, $Z = 3.25$; cluster size = 289), and right cerebellum (MNI = 4, − 78, −36, $Z = 3.60$; cluster size = 1,074) (Fig. 1A). However, we did not observe any GMV predictions in the subdimensional scores of maximizing tendency.

To directly compare the univariate analysis and MVPA results, we plotted the z-score of each voxel in the ROIs. As indicated in Fig. 1B–D, the z-scores of the two analyses were significantly correlated in the IFG ($r = 0.093$, $p = 0.012$; Fig. 1B), insula ($r = 0.304$, $p < 0.001$; Fig. 1C), and cerebellum ($r = -0.143$, $p < 0.001$; Fig. 1D), suggesting that the effect size was lower for univariate analysis than for MVPA.

Discussion

By using both univariate analysis and MVPA, we explored the neural correlates of maximizing tendency, which might involve important brain regions related to the cognitive and emotional components of maximizers. Previous studies have suggested that the prefrontal cortex, medial OFC, right IFG, and insula play a major role in social comparisons, confidence, and regret experiences. Accordingly, we predicted that GMV data in these brain regions would be associated with individual differences in maximizing tendency. Furthermore, we predicted that such associations would be more likely to be observed in MVPA than in the traditional univariate analysis approach. Indeed, our univariate analysis of VBM did not reveal any significant interaction effects between maximization and sex or main effect of maximization on GMV in the whole brain. However, MVPA of VBM revealed that GMV data in the right IFG, insula, and cerebellum could successfully predict individual differences in maximization. Notably, such patterns were not replicated in the three subcomponents of maximization (i.e., alternative search, decision difficulty, and high standards).

Although our univariate analysis did not show any significant relationships between GMV and maximization, we demonstrate for the first time that MVPA revealed that GMV data in right IFG, insula, and cerebellum could significantly predict individual differences in maximizing tendency. Previous studies have demonstrated that right IFG and insula activities reflect upward social comparison, confidence, and counterfactual thinking (Lindner et al., 2015; Van Hoeck et al., 2012). In particular, downward and upward comparisons exhibited distinct BOLD activity patterns; downward comparisons involved the ventral anterior cingulate cortex (vACC) and medial OFC, and upward comparisons involved the IFG, anterior insula, and the dorsomedial PFC (Lindner et al., 2015). We also found that counterfactual thinking was associated with greater activations in the IFG and cerebellum (Van Hoeck et al., 2012). Thus, maximizing motivates individuals to search for the best option, which eventually leads them to question whether their decisions could have been better. By sustaining a search for better options after completing their decisions, maximizers become increasingly likely to employ some decision strategies such as upward social comparisons and counterfactual thinking that are subserved by the IFG, insula, and cerebellum, which further undermines their satisfaction even when they obtain relatively good enough outcomes (Iyengar et al., 2006; Lai, 2011; Sparks et al., 2012).

When people make decisions, various psychological processes are involved, including error monitoring, response inhibition, decision conflicting, outcome feedbacking, emotional modulation, and cognitive control. As demonstrated in the literature, these specific psychological processes are broadly associated with prefrontal cortex function, especially IFG function. In particular, previous studies have determined that IFG activation was sensitive to no-go suppression (Aron et al., 2004) and that the volume of the IFG varied with response inhibition ability estimated using the stop–signal response time (Aron et al., 2003). Empirical studies have indicated that the modulation of the IFG on value computation may be related to several available alternatives (Dixon et al., 2017), emotional experiences formed after decisions have been made (Herrington et al., 2005), flexible and goal-directed behavior reflecting maximization (Verbruggen & Logan, 2008), and the implementation of control (MacDonald et al., 2000). Because task-based fMRI studies on maximizing are scarce, the specific function of the aforementioned brain regions still warrants further empirical investigation.

In addition to the IFG, we found that the GMV of the insula was associated with maximizing tendency at the individual level. The insular cortex is an integral brain hub connecting different functional systems underlying sensory, emotional, motivational, and cognitive processing (Gogolla, 2017). Previous research has demonstrated that maximizers seek the "best" option across decisions and report lower life satisfaction and happiness and greater regret and depression (Iyengar et al., 2006; Polman, 2010). The insula is involved in processing emotional experiences (Gogolla, 2017) and exhibits altered functioning and structural characteristics across different forms of anxiety disorders (Kawaguchi et al., 2016). A meta-analysis indicated that increased GMV in the insula was associated with mental disorders (Carlisi et al., 2017). Furthermore, the insula was found to project to brain regions implicated in motivation and reward, such as the nucleus accumbens and the caudate putamen (Gogolla, 2017); feelings of regret also activated the anterior insula (Chua et al., 2009). Overall, in addition to its cognitive role, the insula might be involved in negative emotional experiences in individuals' thought processes when maximizing their rewards.

Neither behavioral nor univariate imaging findings reveal sex differences in maximizing tendency; these findings are consistent with those of previous research (Richardson et al., 2014). Sex may be modulated in decision-making; nevertheless, the present study suggested that individual differences in maximizing tendency had similar distributions between male and female participants, as did the brain structural organizations related to maximizing. Considering the sex differences in social comparisons and regret (Guimond et al., 2006; Roese et al., 2006), we believe that maximizing is a complex process that involves separate mental components that cannot be interpreted on the basis of only social comparisons and regret.

The present study has limitations. First, our participants were healthy Chinese students; this thus limits the generalizability of our findings. Additionally, previous studies have observed that maximizing tendency is negatively correlated with age (Bruine de Bruin et al., 2016), which could not be replicated in the present study, possibly due to the relatively narrow age range of the participants. Second, our study's data could not describe causality. Accordingly, the present study could not determine whether the increase in the volume of the right IFG affected maximizing tendency or whether maximizing tendency caused the increase in the volume of the right IFG. Third, although previous studies have demonstrated similar statistical power and consistent findings between k-fold cross-validation and leave-one-out strategy (Fushiki, 2011; Li et al., 2007), the cross-validation error can be affected by the type of cross-validation, the present study only used 3-fold cross-validation strategy.

Conclusions

The morphological characteristics of the cortex, insula, and cerebellum determine maximizing tendency, which manifests similarly in male and female individuals. Such brain structural organizations suggest the involvement of psychological processes, such as upward social comparisons and counterfactual thinking, in maximizing.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s11682-022-00656-3.

Authors' contributions Author contributions included data collection, acquisition, and processing (HZ, SW, YW, JF), statistical analysis (HZ, JF), interpretation of results (HZ, JF), drafting the manuscript work or revising it critically for important intellectual content (HZ, JF), and approval of final version to be published and agreement to be accountable for the integrity and accuracy of all aspects of the work (all authors).

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Declarations

Ethics approval This study was approved by the Institutional Review Board of the School of Psychological and Cognitive Sciences at Peking University and Faculty of Psychology at Tianjin Normal University.

Consent to participate Informed written consent was obtained from participants before formal experiments were conducted.

Conflict of interest All authors declare that they have no conflict of interest to declare.

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